

ICT702 Data Visualisation

Explaining Visually to Influence with Data Module 7





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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Learning Objectives for Module 7

- Explain the importance of knowing your audience's needs and analytical comfort level to create an effective data visualisation or presentation
- Explain how to create empathy in the audience with the data to create the most effective message possible in your data visualisation or presentation
- List the types of data visualisations that are most appropriate to communicate specific insights and for audiences with different needs and different levels of analytical comfort
- Use dot matrix charts and big associated numbers to give the audience a better relative understanding of large numerical values in a data visualisation
- Describe examples of how data visualisations can be created to help the audience empathize with data

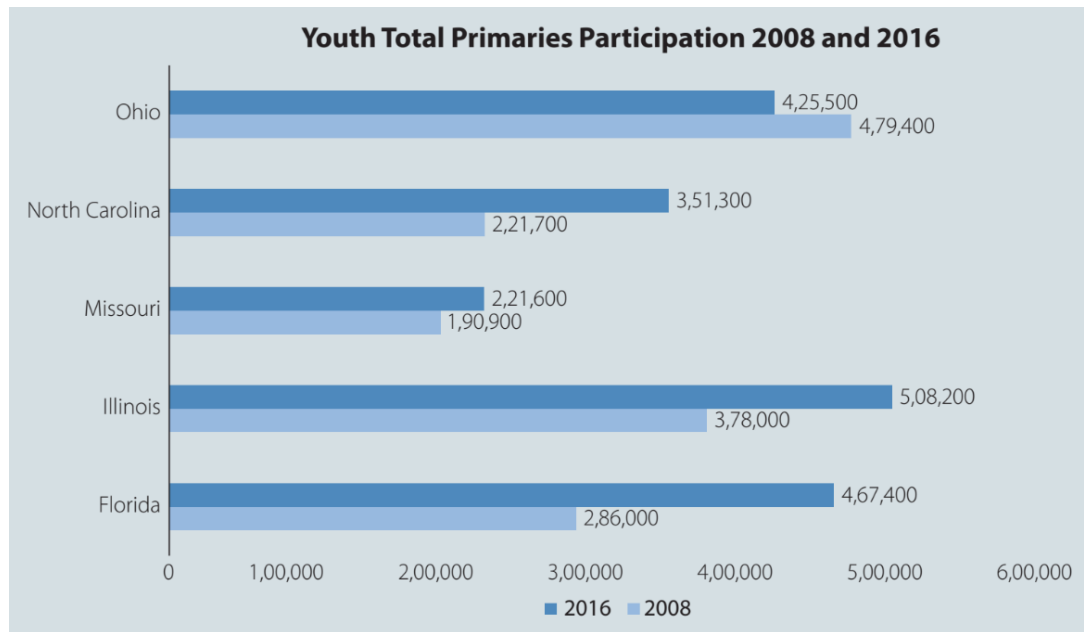
Learning Objectives for Module 7 cont.

- Identify situations for which a slope chart is appropriate and be able to create a slope chart from data
- Use preattentive attributes to emphasize certain insights for an audience and tell a story in a data visualisation
- Define Aristotle's Rhetorical Triangle and explain how it can be employed to connect with the audience to tell a story in a data visualisation or presentation
- Define Freytag's Pyramid and explain how this provides a suggested structure for telling a story in your presentation
- Define the concept of storyboarding and explain how to storyboard a presentation using sticky notes and using PowerPoint
- Create an effective story for a presentation that considers the needs of the audience

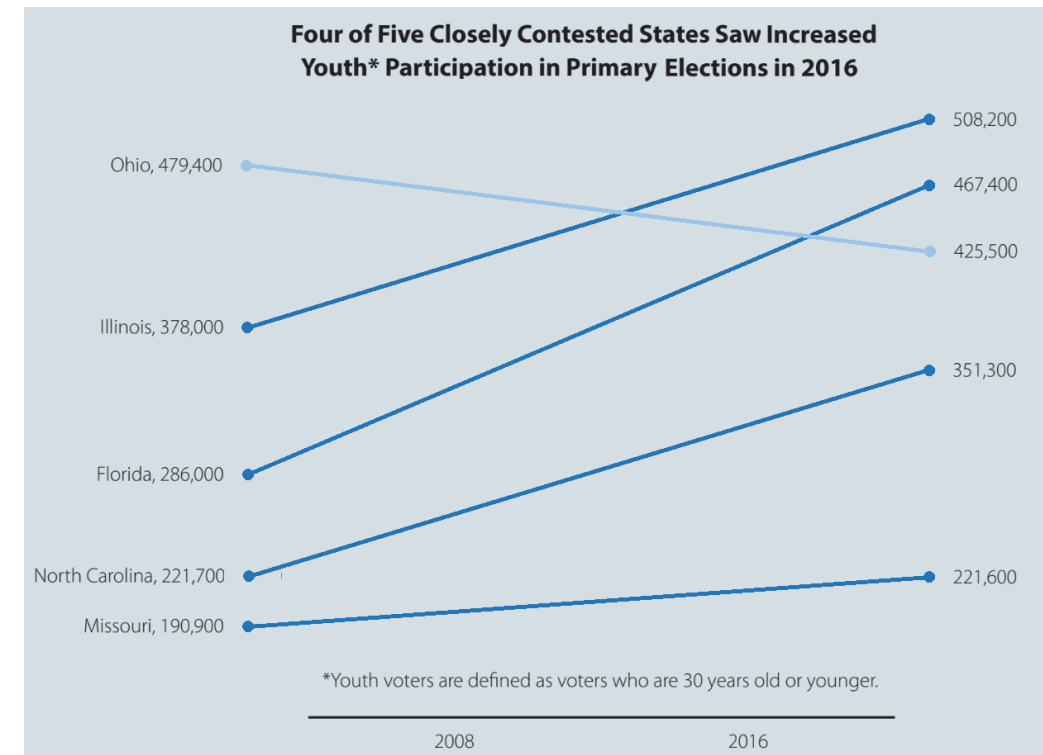
Part 1

Data Visualisation Makeover

Clustered Column Chart



Slope Chart



Storytelling

Storytelling refers to the ability to build a narrative from the data that is meaningful and memorable and is likely to influence the audience.

The first goal in effective storytelling is to ensure we understand our audience by determining:

1. The insight needs of the audience from our data visualisation:
 - High-level understanding
 - Detailed understanding
2. The level of analytical comfort in our audience:
 - Low analytical comfort
 - High analytical comfort

Audience Member Needs

The Chan Life Insurance Group investment in new process technology

Task	Average Time to Complete Task (minutes)					
	January	March	May	July	September (Projected)	November (Projected)
Process Request for Quote	244	230	267	220	70	50
Underwrite and Generate Quote	154	167	172	168	40	20
Verify Quote	98	112	110	115	20	10
Send Quote to Agent	121	115	110	117	120	120
Total Response Time	617	624	659	620	250	200

Chan Life Insurance wants to provide its employees with updates on the new process.

Two major audiences of employees with different needs:

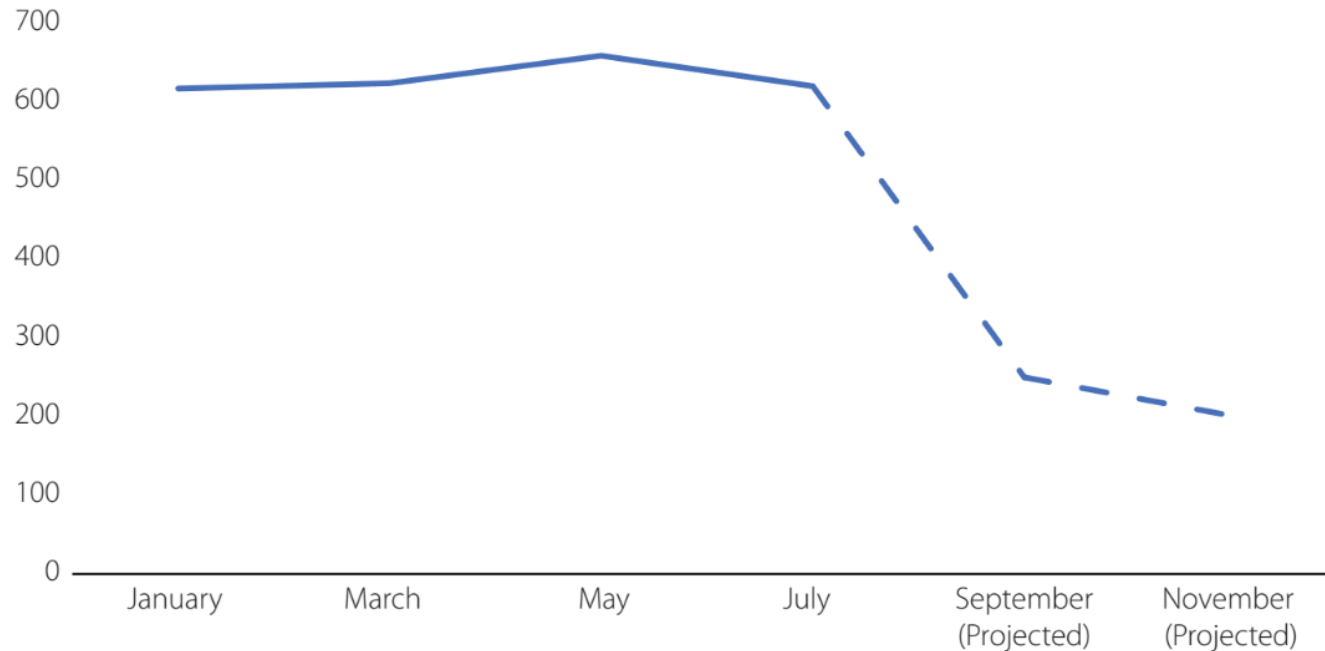
- Sales agents' needs
- Underwriters' needs

Need for High-Level Understanding

Line chart for displaying the total response time for quotes

New IT System Projected to Decrease Response Time for Quote Requests

Total Response Time (minutes)



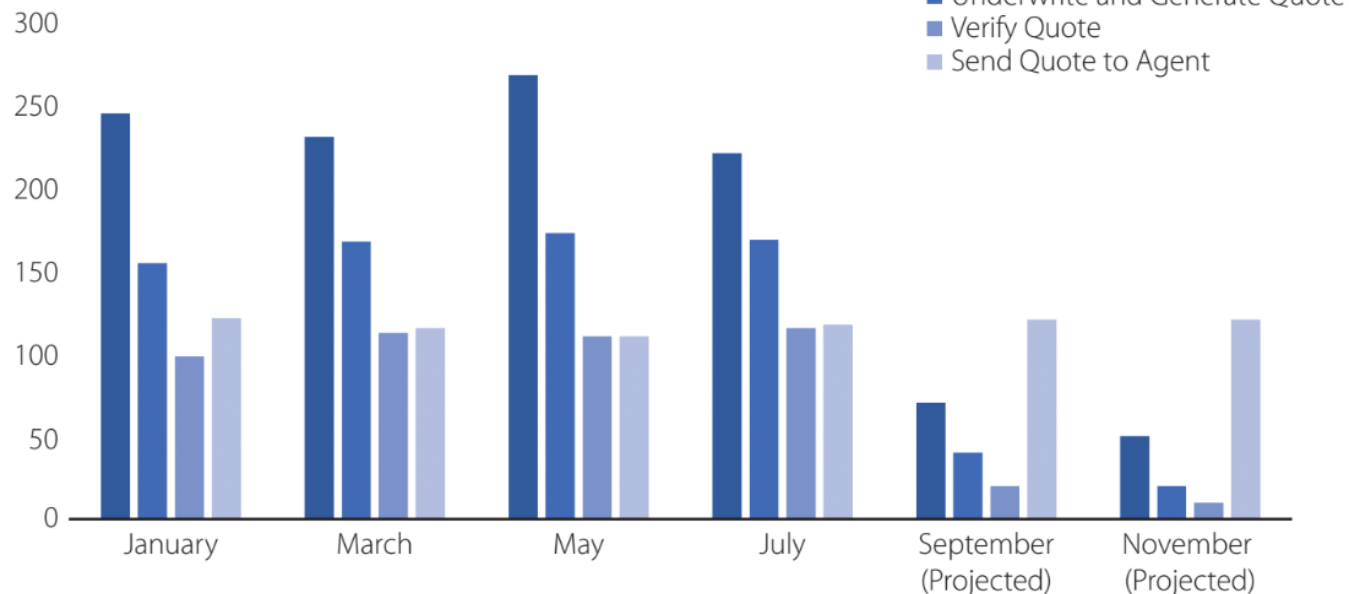
Sales agents → high-level understanding
September and November projections are indicated with a dashed line.

Need for Detailed Understanding

Clustered column chart for displaying the task times for responses to quotes

New IT System Projected to Decrease Processing, Underwriting, and Verification Times

Average Time to Complete Task (minutes)



Underwriters → detailed understanding

Chart may warrant further conversations on impact of new process technology on specific tasks.

Audience Member Analytical Comfort Levels

Patient satisfaction survey results at General Hospital

	A	B	C	D	E	F	G	H
1	Patient Satisfaction Scores							
2	Neurologic	General	Vascular	Oncology	Pediatric	Ophthalmic	Thoracic	Orthopaedic
3	1.9	3.6	3.0	2.9	3.3	3.9	4.5	4.4
4	2.8	2.5	3.5	3.7	4.6	4.0	5.0	4.5
5	2.1	2.1	1.5	3.4	3.9	4.1	4.7	4.5
6	3.2	2.8	2.6	3.8	4.1	4.0	5.0	4.4
7	2.3	3.8	1.5	3.3	3.1	3.8	4.9	4.5
8	2.7	2.6	2.2	4.3	2.9	4.2	4.4	4.6
9	2.9	3.9	2.5	3.2	4.0	3.9	4.1	4.4
10	2.0	2.4	3.1	3.2	4.6	4.1	5.0	4.4
11	2.6	2.5	2.3	4.3	4.1	4.0	4.7	4.5
12	1.6	1.6	3.1	3.5	3.8	4.1	4.5	4.5

General Hospital presents regular updates on patient satisfaction survey results from surgical patients to its staff.

Two main audiences with either a low or a high analytical comfort level.

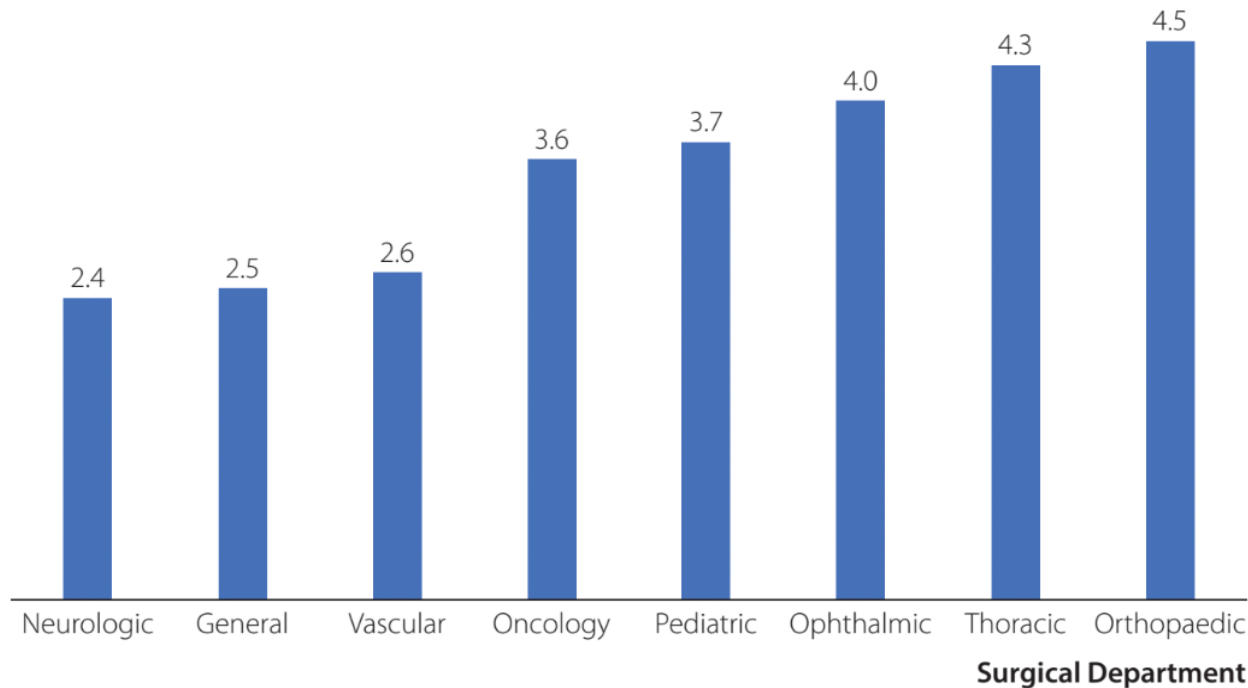


Low Analytical Comfort Level

Column chart displaying average patient satisfaction scores for surgical patients

Average Patient Satisfaction Scores at General Hospital

Average Score (0 = Very Dissatisfied; 5 = Very Satisfied)



Low analytical comfort level

- Provides simple overview of the main results
- Makes relative comparisons among the department easier

High Analytical Comfort Level

Box and whisker chart displaying details on patient satisfaction scores for surgical patients

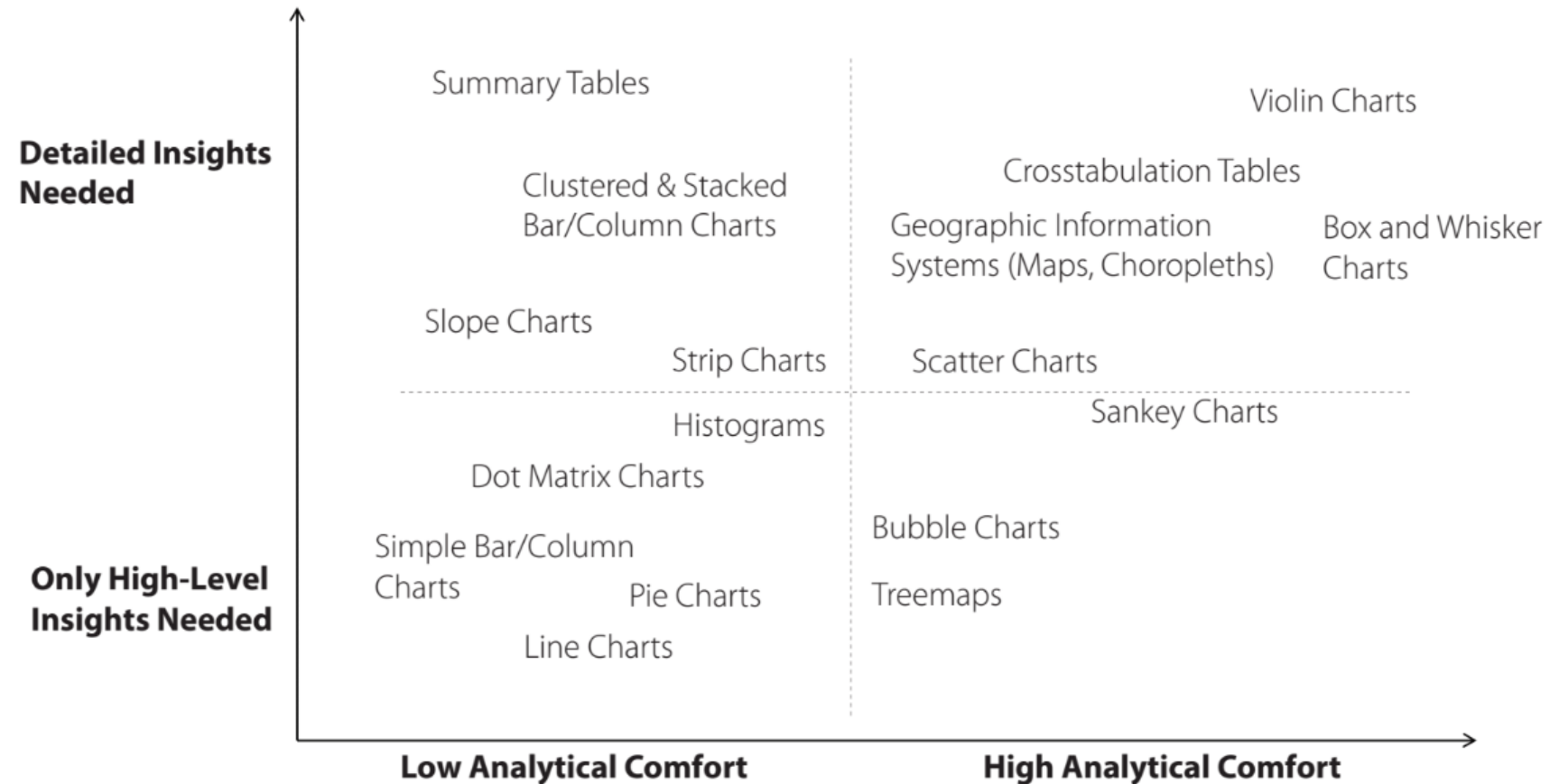


High analytical comfort level

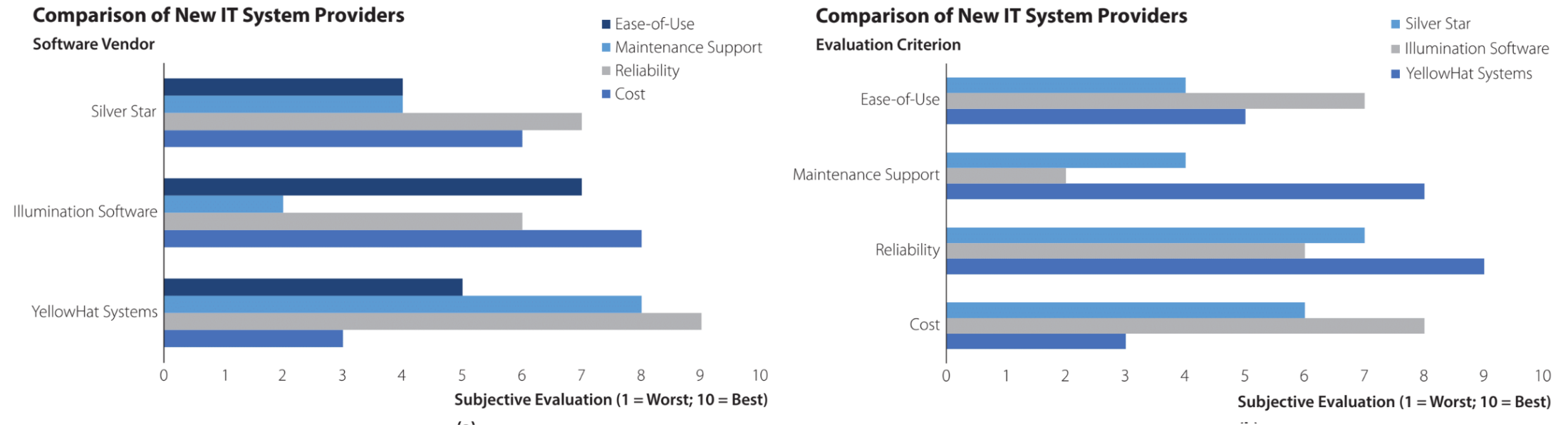
- Provides more insights into the results of the patient satisfaction scores

Best Choice of Data Visualisation

Depends on the
Needs and
Analytical comfort
Level of the
Intended Audience



What Helps the Decision-Maker?



The two different clustered bar charts assist a decision-maker choosing among three competing new IT systems.

End of Part 1